Code No: 134AK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2018
COMPUTER ORGANIZATION
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) Write the generic Instruction types present in a computer system. [2]
b) What is the difference between a direct and an indirect address instruction? [3]
c) List the four basic functions of the CPU. [2]
d) Give a note on Instruction Set of 8086. [3]
e) What is an interrupt service routine in microprocessor? [2]
f) How a clock signal is generated in 8086 microprocessor? [3]
g) List four peripherals devices that produce an acceptable output for a person to understand. [2]
h) How many characters per second can be transmitted over a 1200 baud line in Synchronous serial transmission? [3]
i) What are the difficulties that cause the instruction pipeline to deviate from its normal operation? [2]
j) Draw the structure of general purpose multicomputer. [3]

PART - B

5 × 10 marks = 50

2.a) How many references to memory are needed for each type of instruction to bring an operand into a processor register? Explain.
b) With the help of a block diagram, explain how do we select the address of control memory. [5+5]

OR

3.a) Give a brief note on instruction cycle.
b) List and explain the functional units of a computer. [5+5]

4. Draw and explain the 8086 Processor Architecture. [10]

OR

5.a) Explain the Assembler Directives**.**
b) Discuss the Physical memory organization. [5+5]

6. How to pass parameters to procedures in 8086? Explain in detail with an ALP**.** [10]

OR

7.a) Is ‘c’ an assembly language? Justify your answer.
b) With an assembly language program explain stack organization in 8086. [4+6]

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8. Compare interrupt driven data transfer scheme with DMA. Using block diagram explain interrupt driven transfer scheme. [10]

OR

9. Explain Booths multiplication algorithm with example. [10]

10.a) Distinguish between the virtual memory and cache memory. Write the merits and demerits of virtual memory.

b) Give a neat sketch that illustrates the components in a typical memory hierarchy. [5+5]

OR

11.a) With the help of a neat diagram explain the match logic for one word of associative memory.

b) What are the various forms available for establishing an interconnection network in a multi processor system? [5+5]

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